



Parshvanath Charitable Trust's
A. P. SHAH INSTITUTE OF TECHNOLOGY
(Approved by AICTE New Delhi & Govt. of Maharashtra, Affiliated to University of Mumbai)
(Religious Jain Minority)



Academic Year: 2025-26
Semester: V
Class : TE (Div-C)
Subject: Software Engineering Lab(CSL 601)

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Date of Performance: 07/10/25
Date of Submission: 09/10/25

Name of Instructor: **Prof. Manasi Chouk**

Batch: C3

Experiment 10

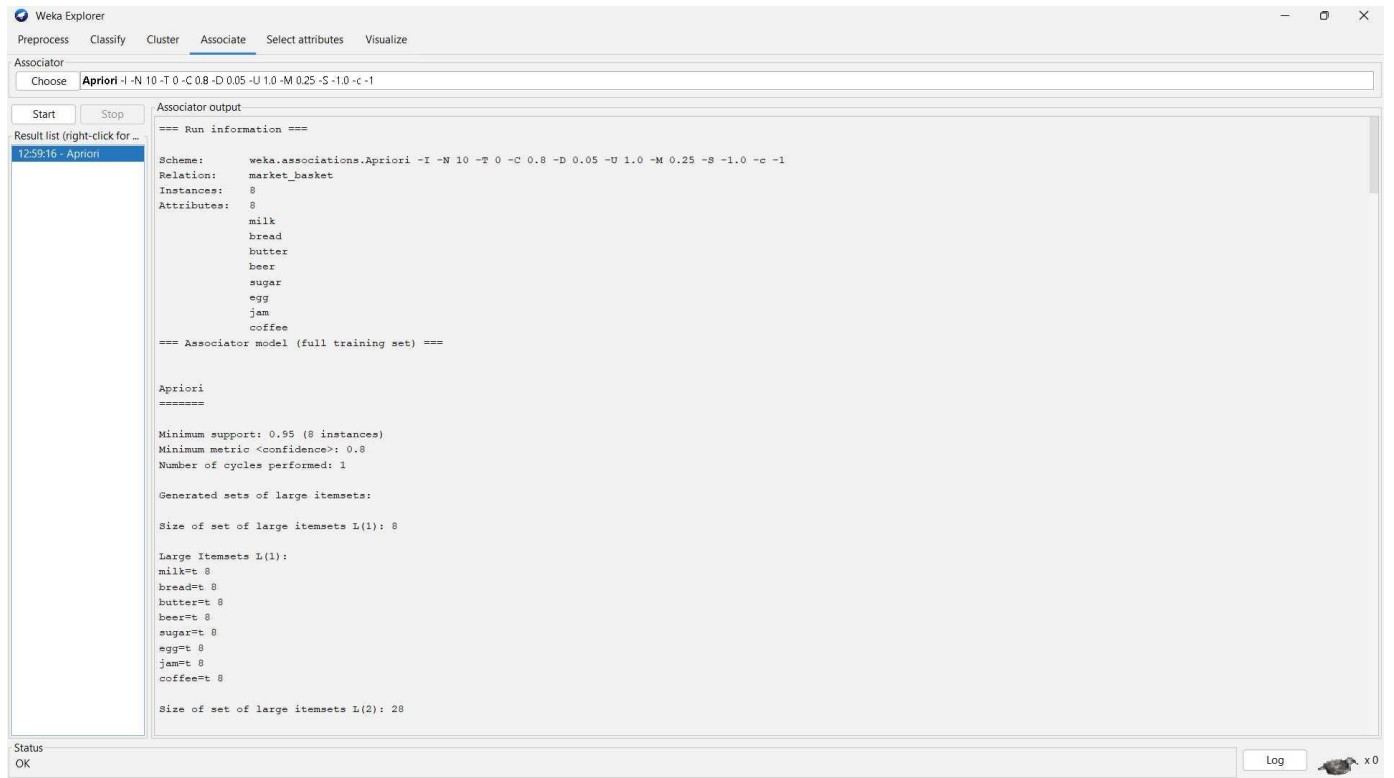
Aim: Implementation of the Association Rule Mining algorithm (Apriori)

Program:

```
@relation market_basket
@attribute milk {t}
@attribute bread {t}
@attribute butter {t}
@attribute beer {t}
@attribute sugar {t}
@attribute egg {t}
@attribute jam {t}
@attribute coffee {t}
```

```
@data
{0 t, 1 t, 2 t, 3 t}
{0 t, 1 t, 4 t, 5 t}
{0 t, 1 t, 2 t, 6 t}
{0 t, 1 t, 4 t}
{4 t, 5 t, 7 t}
{0 t, 1 t, 6 t}
{2 t, 4 t, 5 t, 7 t}
{0 t, 5 t, 7 t}
```

Output:



```
Weka Explorer
Preprocess  Classify  Cluster  Associate  Select attributes  Visualize

Associator
Choose  Apriori -I -N 10 -T 0 -C 0.8 -D 0.05 -U 1.0 -M 0.25 -S -1.0 -c -1

Start  Stop
Result list (right-click for ...)
12:59:16 - Apriori

Associator output

=== Run information ===

Scheme:      weka.associations.Apriori -I -N 10 -T 0 -C 0.8 -D 0.05 -U 1.0 -M 0.25 -S -1.0 -c -1
Relation:    market_basket
Instances:   8
Attributes:  8
             milk
             bread
             butter
             beer
             sugar
             egg
             jam
             coffee

=== Associator model (full training set) ===

Apriori
=====

Minimum support: 0.95 (8 instances)
Minimum metric <confidence>: 0.8
Number of cycles performed: 1

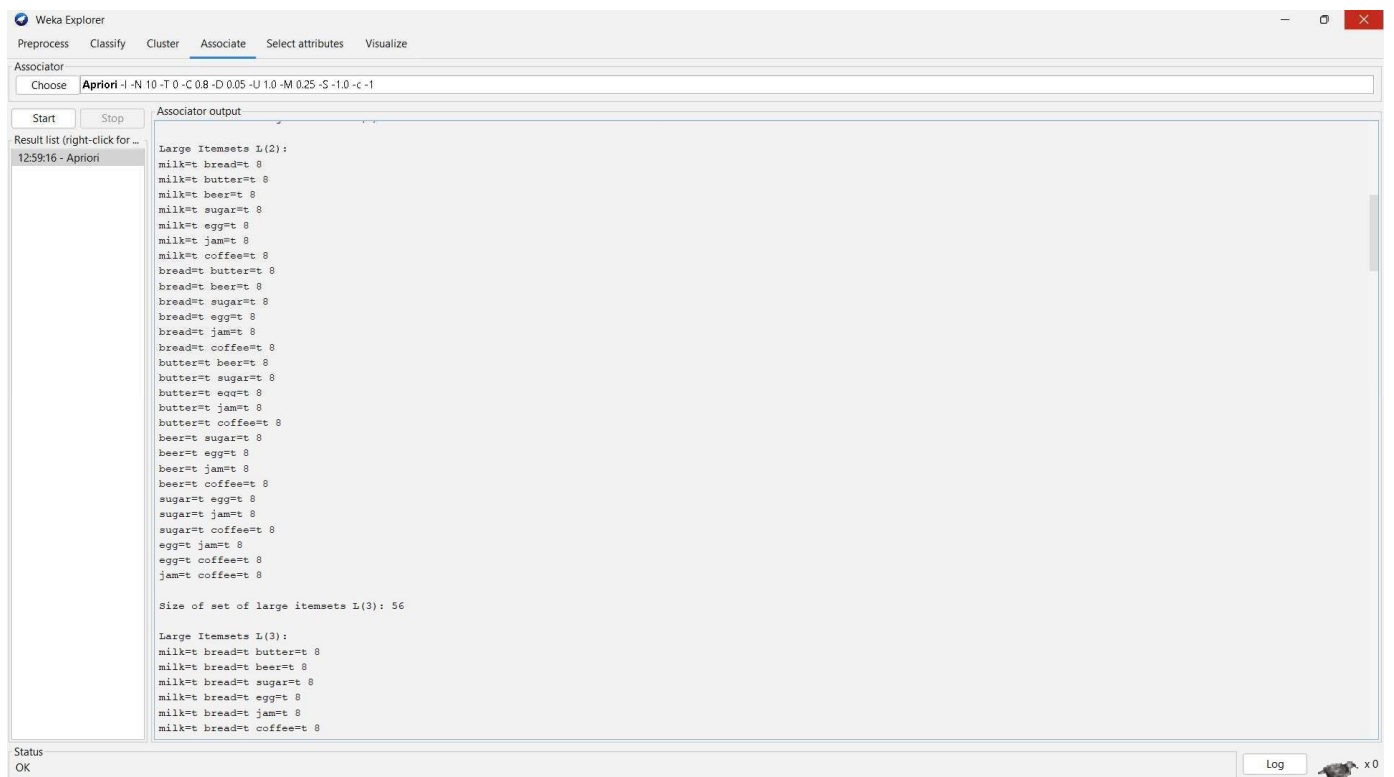
Generated sets of large itemsets:

Size of set of large itemsets L(1): 8

Large Itemsets L(1):
milk=t 8
bread=t 8
butter=t 8
beer=t 8
sugar=t 8
egg=t 8
jam=t 8
coffee=t 8

Size of set of large itemsets L(2): 28

Status
OK
```



```
Weka Explorer
Preprocess  Classify  Cluster  Associate  Select attributes  Visualize

Associator
Choose  Apriori -I -N 10 -T 0 -C 0.8 -D 0.05 -U 1.0 -M 0.25 -S -1.0 -c -1

Start  Stop
Result list (right-click for ...)
12:59:16 - Apriori

Associator output

Large Itemsets L(2):
milk=t bread=t 8
milk=t butter=t 8
milk=t beer=t 8
milk=t sugar=t 8
milk=t egg=t 8
milk=t jam=t 8
milk=t coffee=t 8
bread=t butter=t 8
bread=t beer=t 8
bread=t sugar=t 8
bread=t egg=t 8
bread=t jam=t 8
bread=t coffee=t 8
butter=t beer=t 8
butter=t sugar=t 8
butter=t egg=t 8
butter=t jam=t 8
butter=t coffee=t 8
beer=t sugar=t 8
beer=t egg=t 8
beer=t jam=t 8
beer=t coffee=t 8
sugar=t egg=t 8
sugar=t jam=t 8
sugar=t coffee=t 8
egg=t jam=t 8
egg=t coffee=t 8
jam=t coffee=t 8

Size of set of large itemsets L(3): 56

Large Itemsets L(3):
milk=t bread=t butter=t 8
milk=t bread=t beer=t 8
milk=t bread=t sugar=t 8
milk=t bread=t egg=t 8
milk=t bread=t jam=t 8
milk=t bread=t coffee=t 8

Status
OK
```

Weka Explorer

Preprocess Classify Cluster **Associate** Select attributes Visualize

Associator

Choose **Apriori** -I -N 10 -T 0 -C 0.8 -D 0.05 -U 1.0 -M 0.25 -S -1.0 -c -1

Start Stop

Result list (right-click for ...)

12:59:16 - Apriori

Associator output

Size of set of large itemsets L(4): 70

Large Itemsets L(4):

```

milk=t bread=t butter=t beer=t 8
milk=t bread=t butter=t sugar=t 8
milk=t bread=t butter=t egg=t 8
milk=t bread=t butter=t jam=t 8
milk=t bread=t butter=t coffee=t 8
milk=t bread=t beer=t sugar=t 8
milk=t bread=t beer=t egg=t 8
milk=t bread=t beer=t jam=t 8
milk=t bread=t beer=t coffee=t 8
milk=t bread=t sugar=t egg=t 8
milk=t bread=t sugar=t jam=t 8
milk=t bread=t sugar=t coffee=t 8
milk=t bread=t egg=t jam=t 8
milk=t bread=t egg=t coffee=t 8
milk=t butter=t beer=t sugar=t 8
milk=t butter=t beer=t egg=t 8
milk=t butter=t beer=t jam=t 8
milk=t butter=t beer=t coffee=t 8
milk=t butter=t sugar=t egg=t 8
milk=t butter=t sugar=t jam=t 8
milk=t butter=t sugar=t coffee=t 8
milk=t butter=t egg=t jam=t 8
milk=t butter=t egg=t coffee=t 8
milk=t butter=t jam=t coffee=t 8
milk=t beer=t sugar=t egg=t 8
milk=t beer=t sugar=t jam=t 8
milk=t beer=t sugar=t coffee=t 8
milk=t beer=t egg=t jam=t 8
milk=t beer=t egg=t coffee=t 8
milk=t beer=t jam=t coffee=t 8
milk=t sugar=t egg=t jam=t 8
milk=t sugar=t egg=t coffee=t 8
milk=t sugar=t jam=t coffee=t 8
milk=t egg=t jam=t coffee=t 8
bread=t butter=t beer=t sugar=t 8
bread=t butter=t beer=t egg=t 8

```

Status

OK

Log x 0

Weka Explorer

Preprocess Classify Cluster **Associate** Select attributes Visualize

Associator

Choose **Apriori** -I -N 10 -T 0 -C 0.8 -D 0.05 -U 1.0 -M 0.25 -S -1.0 -c -1

Start Stop

Result list (right-click for ...)

12:59:16 - Apriori

Associator output

Size of set of large itemsets L(5): 56

Large Itemsets L(5):

```

milk=t bread=t butter=t beer=t sugar=t 8
milk=t bread=t butter=t beer=t egg=t 8
milk=t bread=t butter=t beer=t jam=t 8
milk=t bread=t butter=t beer=t coffee=t 8
milk=t bread=t butter=t sugar=t egg=t 8
milk=t bread=t butter=t sugar=t jam=t 8
milk=t bread=t butter=t sugar=t coffee=t 8
milk=t bread=t butter=t egg=t jam=t 8
milk=t bread=t butter=t egg=t coffee=t 8
milk=t bread=t butter=t jam=t coffee=t 8
milk=t bread=t beer=t sugar=t egg=t 8
milk=t bread=t beer=t sugar=t jam=t 8
milk=t bread=t beer=t sugar=t coffee=t 8
milk=t bread=t beer=t egg=t jam=t 8
milk=t bread=t beer=t egg=t coffee=t 8
milk=t bread=t beer=t jam=t coffee=t 8
milk=t bread=t sugar=t egg=t jam=t 8
milk=t bread=t sugar=t egg=t coffee=t 8
milk=t bread=t sugar=t jam=t coffee=t 8
milk=t bread=t egg=t jam=t coffee=t 8
milk=t butter=t beer=t sugar=t egg=t 8
milk=t butter=t beer=t sugar=t jam=t 8
milk=t butter=t beer=t sugar=t coffee=t 8
milk=t butter=t beer=t egg=t jam=t 8
milk=t butter=t beer=t egg=t coffee=t 8
milk=t butter=t beer=t jam=t coffee=t 8
milk=t butter=t sugar=t egg=t jam=t 8
milk=t butter=t sugar=t egg=t coffee=t 8
milk=t butter=t sugar=t jam=t coffee=t 8
milk=t butter=t egg=t jam=t coffee=t 8
milk=t butter=t egg=t coffee=t 8
milk=t beer=t sugar=t egg=t jam=t 8
milk=t beer=t sugar=t egg=t coffee=t 8
milk=t beer=t sugar=t jam=t coffee=t 8
milk=t sugar=t egg=t jam=t coffee=t 8
bread=t butter=t beer=t sugar=t egg=t 8
bread=t butter=t beer=t sugar=t jam=t 8

```

Status

OK

Log x 0

Weka Explorer

Preprocess Classify Cluster **Associate** Select attributes Visualize

Associator

Choose **Apriori** -I -N 10 -T 0 -C 0.8 -D 0.05 -U 1.0 -M 0.25 -S -1.0 -c -1

Start Stop

Result list (right-click for ...)

12:59:16 - Apriori

Associator output

```

bread=t beer=t egg=t jam=t coffee=t 8
bread=t sugar=t egg=t jam=t coffee=t 8
butter=t beer=t sugar=t egg=t jam=t 8
butter=t beer=t sugar=t egg=t coffee=t 8
butter=t beer=t sugar=t jam=t coffee=t 8
butter=t beer=t egg=t jam=t coffee=t 8
butter=t sugar=t egg=t jam=t coffee=t 8
beer=t sugar=t egg=t jam=t coffee=t 8

Size of set of large itemsets L(6): 28

Large Itemsets L(6):
milk=t bread=t butter=t beer=t sugar=t egg=t 8
milk=t bread=t butter=t beer=t sugar=t jam=t 8
milk=t bread=t butter=t beer=t sugar=t coffee=t 8
milk=t bread=t butter=t beer=t egg=t jam=t 8
milk=t bread=t butter=t beer=t egg=t coffee=t 8
milk=t bread=t butter=t beer=t jam=t coffee=t 8
milk=t bread=t butter=t sugar=t egg=t jam=t 8
milk=t bread=t butter=t sugar=t egg=t coffee=t 8
milk=t bread=t butter=t sugar=t jam=t coffee=t 8
milk=t bread=t butter=t egg=t jam=t coffee=t 8
milk=t bread=t beer=t sugar=t egg=t jam=t 8
milk=t bread=t beer=t sugar=t egg=t coffee=t 8
milk=t bread=t beer=t sugar=t jam=t coffee=t 8
milk=t bread=t beer=t egg=t jam=t coffee=t 8
milk=t bread=t sugar=t egg=t jam=t coffee=t 8
milk=t butter=t beer=t sugar=t egg=t jam=t 8
milk=t butter=t beer=t sugar=t egg=t coffee=t 8
milk=t butter=t beer=t sugar=t jam=t coffee=t 8
milk=t butter=t beer=t egg=t jam=t coffee=t 8
milk=t butter=t sugar=t egg=t jam=t coffee=t 8
milk=t beer=t sugar=t egg=t jam=t coffee=t 8
bread=t butter=t beer=t sugar=t egg=t jam=t 8
bread=t butter=t beer=t sugar=t egg=t coffee=t 8
bread=t butter=t beer=t sugar=t jam=t coffee=t 8
bread=t butter=t beer=t egg=t jam=t coffee=t 8
bread=t butter=t sugar=t egg=t jam=t coffee=t 8
bread=t beer=t sugar=t egg=t jam=t coffee=t 8
butter=t beer=t sugar=t egg=t jam=t coffee=t 8

```

Status OK Log x0

Weka Explorer

Preprocess Classify Cluster **Associate** Select attributes Visualize

Associator

Choose **Apriori** -I -N 10 -T 0 -C 0.8 -D 0.05 -U 1.0 -M 0.25 -S -1.0 -c -1

Start Stop

Result list (right-click for ...)

12:59:16 - Apriori

Associator output

```

milk=t beer=t sugar=t egg=t jam=t coffee=t 8
bread=t butter=t beer=t sugar=t egg=t jam=t 8
bread=t butter=t beer=t sugar=t egg=t coffee=t 8
bread=t butter=t beer=t sugar=t jam=t coffee=t 8
bread=t butter=t beer=t egg=t jam=t coffee=t 8
bread=t butter=t sugar=t egg=t jam=t coffee=t 8
bread=t beer=t sugar=t egg=t jam=t coffee=t 8
butter=t beer=t sugar=t egg=t jam=t coffee=t 8

Size of set of large itemsets L(7): 8

Large Itemsets L(7):
milk=t bread=t butter=t beer=t sugar=t egg=t jam=t 8
milk=t bread=t butter=t beer=t sugar=t egg=t coffee=t 8
milk=t bread=t butter=t beer=t sugar=t jam=t coffee=t 8
milk=t bread=t butter=t beer=t egg=t jam=t coffee=t 8
milk=t bread=t butter=t sugar=t egg=t jam=t coffee=t 8
milk=t bread=t beer=t sugar=t egg=t jam=t coffee=t 8
milk=t butter=t beer=t sugar=t egg=t jam=t coffee=t 8
bread=t butter=t beer=t sugar=t egg=t jam=t coffee=t 8

Size of set of large itemsets L(8): 1

Large Itemsets L(8):
milk=t bread=t butter=t beer=t sugar=t egg=t jam=t coffee=t 8

Best rules found:

1. bread=t 8 ==> milk=t 8 <conf: (1)> lift: (1) lev: (0) [0] conv: (0)
2. milk=t 8 ==> bread=t 8 <conf: (1)> lift: (1) lev: (0) [0] conv: (0)
3. butter=t 8 ==> milk=t 8 <conf: (1)> lift: (1) lev: (0) [0] conv: (0)
4. milk=t 8 ==> butter=t 8 <conf: (1)> lift: (1) lev: (0) [0] conv: (0)
5. beer=t 8 ==> milk=t 8 <conf: (1)> lift: (1) lev: (0) [0] conv: (0)
6. milk=t 8 ==> beer=t 8 <conf: (1)> lift: (1) lev: (0) [0] conv: (0)
7. sugar=t 8 ==> milk=t 8 <conf: (1)> lift: (1) lev: (0) [0] conv: (0)
8. milk=t 8 ==> sugar=t 8 <conf: (1)> lift: (1) lev: (0) [0] conv: (0)
9. egg=t 8 ==> milk=t 8 <conf: (1)> lift: (1) lev: (0) [0] conv: (0)
10. milk=t 8 ==> egg=t 8 <conf: (1)> lift: (1) lev: (0) [0] conv: (0)

```

Status OK Log x0

Conclusion:

This lab successfully demonstrates the implementation of the Apriori algorithm for association rule mining using Weka. By setting minimum support and confidence thresholds, we can extract meaningful rules from transactional data, such as {coffee} -> {egg}. These rules provide actionable business insights, like optimizing product placement or creating bundle promotions. The exercises further show that while Apriori is a foundational algorithm, FP-Growth offers a more efficient and scalable alternative by constructing a compact FP-tree, thereby avoiding the costly candidate generation step and reducing database scans.

Exercise 1:

```

Command Prompt
Successfully installed contourpy-1.3.3 cycler-0.12.1 fonttools-4.60.1 kiwisolver-1.4.9 matplotlib-3.10.6 mltend-0.23.4 packaging-25.0 pillow-11.3.0 pyparsing-3.2.5
C:\Users\HP>cd Documents
C:\Users\HP\Documents>python fp_growth_exercise.py
'python' is not recognized as an internal or external command,
operable program or batch file.
C:\Users\HP\Documents>py fp_growth_exercise.py
Original Transactions:
['milk', 'bread', 'butter', 'beer']
['milk', 'sugar', 'egg', 'bread']
['milk', 'bread', 'butter', 'jam']
['bread', 'milk', 'sugar']
['coffee', 'sugar', 'egg']
['milk', 'bread', 'jam']
['sugar', 'coffee', 'egg', 'butter']
['milk', 'coffee', 'egg']

One-Hot Encoded DataFrame (for FP-Growth input):
  beer bread butter coffee egg jam milk sugar
0  True  True   True  False False False  True  False
1  False True  False  False True  False True  True
2  False True   True  False False True  True  False
3  False True  False  False False False True  True
4  False False False  True  True  False False  True
5  False True  False  False False True  True  False
6  False False True   True True  False False  True
7  False False False  True  True  False True  False

Finding Frequent Itemsets with min_support = 0.5 using FP-Growth:
support itemsets
0  0.750 (milk)
1  0.625 (bread)
2  0.500 (sugar)
3  0.500 (egg)
4  0.625 (milk, bread)

Generating Association Rules with min_confidence = 0.7:
antecedents consequents antecedent support consequent support support ... conviction zhangs_metric jaccard certainty kulczynski
0 (bread) (milk) 0.625 0.750 0.625 ... inf 0.666667 0.833333 1.000000 0.916667
1 (milk) (bread) 0.750 0.625 0.625 ... 2.25 1.000000 0.833333 0.555556 0.916667

[2 rows x 14 columns]

--- Interpretation of Association Rules (from FP-Growth) ---
Rule: ['bread'] -> ['milk']
Support: 0.6250
Confidence: 1.0000
Lift: 1.3333

Rule: ['milk'] -> ['bread']
Support: 0.6250

```

Exercise 2:

Feature	Apriori	FP-Growth
Core Strategy	Candidate generation and test	Pattern growth (tree-based)
Database Scans	Multiple (one per itemset length)	Two (fixed)
Candidate Generation	Explicit, can be very large	None (implicit in tree structure)
Memory Usage	Can be high (for candidates)	Generally lower (compact tree), but can be high for very dense data
Speed/Efficiency	Slower, especially for dense/large data	Faster, especially for dense/large data
Scalability	Less scalable for large, dense datasets	More scalable for large, dense datasets
Implementation	Simpler logic to implement from scratch	More complex logic to implement from scratch
Parallelization	Easier to parallelize certain steps	More complex to parallelize tree mining